

Bitcoin Market Sentiment vs Trader Performance Analysis

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Submitted To

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Executive Summary

This project analyzes the relationship between Bitcoin market sentiment and trader performance using Hyperliquid historical trading data and the Bitcoin Fear & Greed Index dataset.

The primary objective of the analysis is to identify sentiment-driven behavioral patterns, profitability trends, and trading risk characteristics within cryptocurrency markets. By combining market sentiment data with historical trading activity, the project investigates how Fear, Greed, Extreme Fear, and Extreme Greed conditions influence trader profitability, market participation, trading direction, and overall risk exposure.

The analysis demonstrates that market sentiment has a measurable impact on trading behavior and profitability distribution. Fear sentiment dominated overall trading activity, while Extreme Greed periods generated the highest average profitability. However, emotionally driven market conditions also introduced significantly higher volatility and downside risk.

The findings further reveal that profitable trading activity is concentrated among a relatively small subset of traders, highlighting the importance of disciplined execution and effective risk management strategies. Overall, the project illustrates how sentiment-aware analytics can improve trading intelligence, support better decision-making, and strengthen market risk controls.

1. Introduction

Cryptocurrency markets are highly volatile and strongly influenced by investor psychology and collective market sentiment. Emotional trading behavior often impacts market participation, position sizing, risk exposure, and profitability outcomes.

The Bitcoin Fear & Greed Index is widely used to measure market emotion by classifying market conditions into categories such as Extreme Fear, Fear, Neutral, Greed, and Extreme Greed. These sentiment indicators provide valuable insights into overall market psychology and potential trader behavior.

This project investigates the relationship between Bitcoin market sentiment and trader performance using historical trading data from Hyperliquid and sentiment classification data from the Fear & Greed Index. The analysis aims to uncover behavioral trading patterns, evaluate profitability across different sentiment conditions, and identify actionable insights that can support smarter trading strategies and improved risk management systems.

2. Objectives

The main objectives of this project are:

1. Analyze the impact of market sentiment on trading behavior.
2. Evaluate trader profitability under different market conditions.
3. Identify behavioral patterns during Fear and Greed phases.
4. Study profitability volatility and risk exposure.
5. Generate actionable insights for trading strategy optimization.
6. Develop sentiment-aware trading intelligence recommendations.

3. Dataset Description

3.1 Bitcoin Fear & Greed Index Dataset

The Fear & Greed Index dataset contains historical Bitcoin market sentiment classifications.

Important Columns

- date
- classification
- value
- timestamp

Sentiment Categories

- Extreme Fear
- Fear
- Neutral
- Greed
- Extreme Greed

The dataset is used to measure emotional market conditions and understand how sentiment influences trading activity and profitability.

3.2 Hyperliquid Historical Trading Dataset

The Hyperliquid dataset contains historical trader execution data and account-level trading activity.

Important Columns

- Account
- Coin
- Execution Price
- Size USD
- Side
- Timestamp IST

- Closed PnL
- Direction

The dataset provides detailed information regarding trader behavior, trade execution, profitability, and trading activity across multiple cryptocurrency assets.

4. Data Preprocessing

Before analysis, both datasets underwent several preprocessing and cleaning operations to ensure accurate integration and reliable analytical results.

Preprocessing Steps Performed

1. Datetime standardization using Pandas datetime utilities.
2. Creation of normalized date columns for dataset merging.
3. Conversion of numerical columns such as:
 - Closed PnL
 - Execution Price
 - Size USD
4. Handling missing values and datatype inconsistencies.
5. Merging the trading dataset with the Fear & Greed Index dataset using normalized date fields.
6. Feature engineering for profitability and trade behavior analysis.

The preprocessing stage ensured accurate alignment between trading activity and corresponding market sentiment conditions.

5. Feature Engineering

Additional features were created to improve behavioral and profitability analysis.

Features Created

Profitability Flag

Trades were classified as profitable or loss-making based on Closed PnL values.

Trade Size Categorization

Trade sizes were grouped into:

- Small
- Medium
- Large

Absolute PnL

Absolute profit/loss values were generated to analyze profitability volatility and risk magnitude.

These engineered features supported deeper behavioral and statistical analysis.

6. Exploratory Data Analysis

6.1 Market Sentiment Distribution

The sentiment distribution analysis examined trading activity across different market sentiment categories.

Observation

Fear sentiment dominated the dataset with the highest trading activity, while Extreme Fear represented the lowest participation levels.

Interpretation

Traders remained actively engaged even during cautious market conditions, indicating sustained market participation despite negative sentiment.

Business Impact

Trading platforms may continue experiencing substantial liquidity and transaction volume during fear-driven markets.

6.2 Average Closed PnL by Sentiment

This analysis evaluated average trader profitability across different market sentiment conditions.

Observation

Extreme Greed periods generated the highest average profitability among all sentiment categories.

Interpretation

Bullish market momentum likely creates stronger trend-following opportunities and increases profitability potential for active traders.

Business Impact

Momentum-based trading strategies may perform more effectively during highly optimistic market environments.

6.3 PnL Distribution Across Market Sentiments

Boxplot analysis was used to study profitability volatility and risk distribution.

Observation

Extreme sentiment conditions displayed significantly larger outliers and profitability dispersion.

Interpretation

Emotionally driven markets amplify both reward potential and downside risk exposure.

Business Impact

Risk management systems should dynamically adjust trading exposure during highly emotional market conditions.

6.4 Buy vs Sell Activity Analysis

The analysis compared buy-side and sell-side participation across different sentiment conditions.

Observation

Sell-side activity slightly exceeded buy-side participation across most market conditions.

Interpretation

Traders may engage in defensive positioning, hedging, or short-term profit-taking behavior during volatile periods.

Business Impact

Directional market imbalance may contribute to increased market volatility and liquidation pressure.

6.5 Top Trader Profitability Analysis

The analysis identified the highest-performing trader accounts based on total Closed PnL.

Observation

A relatively small subset of traders generated disproportionately high profitability.

Interpretation

Consistent profitability appears concentrated among disciplined traders with stronger execution and risk management practices.

Business Impact

Behavioral analytics can help identify high-performing trading strategies and support intelligent copy-trading systems.

6.6 Win Rate Analysis

Trader win rates were analyzed to evaluate profitability consistency.

Observation

Only a limited number of traders consistently maintained high win rates.

Interpretation

Sustainable profitability in crypto trading requires disciplined strategy execution and controlled risk exposure.

Business Impact

Win rate analysis can support trader performance profiling and behavioral risk assessment systems.

6.7 Correlation Analysis

A correlation heatmap was generated to examine relationships between numerical variables such as:

- Execution Price
- Size USD
- Closed PnL

Observation

Trade size and profitability exhibited measurable relationships within the dataset.

Interpretation

Larger positions increase both reward opportunity and downside risk exposure.

Business Impact

Position-sizing controls are critical for maintaining sustainable trading performance.

6.8 Coin-wise Profitability Analysis

Profitability analysis was performed across different cryptocurrency assets.

Observation

Certain cryptocurrency assets demonstrated stronger average profitability compared to others.

Interpretation

Asset-specific volatility and liquidity conditions may influence trading profitability outcomes.

Business Impact

Coin-level analytics can improve asset selection and portfolio optimization strategies.

7. Key Findings

1. Fear sentiment dominated overall market participation.
2. Extreme Greed periods generated the highest average profitability.
3. Emotional market conditions increased profitability volatility and downside risk.
4. Trader profitability was heavily concentrated among a limited subset of accounts.
5. Larger trade sizes demonstrated increased profitability variability.
6. Sell-side activity slightly exceeded buy-side participation across most market conditions.
7. Consistently profitable traders exhibited more disciplined trading behavior and controlled risk exposure.

8. Strategic Recommendations

Based on the analysis, the following strategic recommendations are proposed:

1. Implement sentiment-aware risk management systems.
2. Reduce leverage exposure during Extreme Greed conditions.
3. Monitor trader concentration and behavioral trading risk.
4. Use market sentiment indicators alongside execution strategies.
5. Improve volatility controls during emotionally driven market conditions.
6. Develop trader-behavior analytics to identify disciplined trading patterns.
7. Utilize profitability consistency metrics for trader performance evaluation.

10. Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

9. Conclusion

This analysis demonstrates a strong relationship between market sentiment and trader behavior within cryptocurrency markets. Fear and Greed conditions significantly influence trading activity, profitability distribution, directional trading patterns, and overall market risk exposure.

Extreme Greed periods exhibited the highest profitability potential but also introduced elevated volatility and downside risk, indicating that emotionally driven markets can amplify both gains and losses. Additionally, profitable trading behavior was concentrated among a relatively small group of disciplined traders, emphasizing the importance of effective risk management and controlled execution strategies.

Overall, the project highlights the value of integrating sentiment-aware analytics into trading systems to improve decision-making quality, strengthen risk controls, and support more adaptive trading strategies within highly volatile crypto markets.

11. Future Improvements

Future enhancements for this project may include:

1. Machine learning-based profitability prediction.
2. Trader clustering and behavioral segmentation.
3. Real-time sentiment-driven trading dashboards.
4. Risk scoring systems for trader profiling.
5. Advanced time-series forecasting models.
6. Streamlit-based interactive analytics dashboard.

12. References

1. Bitcoin Fear & Greed Index Dataset
2. Hyperliquid Historical Trading Data
3. Pandas Documentation
4. Matplotlib Documentation
5. Seaborn Documentation
6. Cryptocurrency Market Sentiment Research Papers

